RUDRA PRATAP SINGH

AI and Full-Stack Developer | RAG, Agentic AI, LLM Evaluation

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SUMMARY

Computer Science undergraduate (VTU, CGPA 8.0) building Generative AI applications with Large Language Models (LLMs). Shipped two live, evaluated projects: a multilingual RAG assistant (Python, FastAPI; 0.99 faithfulness) and an agentic AI assistant (Node.js; 20/20 eval accuracy). Founding Software Engineer Intern at FERMOR; seeking AI and full-stack internships.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering (CSE) 2024 - 2028 (Expected) | 5th semester
Cambridge Institute of Technology, North Campus, Bengaluru | Visvesvaraya Technological University (VTU) Current CGPA: 8.0 / 10

Class XII (CBSE) | Physics, Chemistry and Mathematics (PCM) 74%

Class X (CBSE) 74%

EXPERIENCE

Founding Software Engineer Intern, FERMOR Sep 2026 - Oct 2026
Bengaluru, Karnataka (Hybrid)

- Redesigned key website pages end to end, delivering a cleaner and more usable UI/UX with a modern frosted-glass design.
- Optimised page structure, headings, metadata and content for SEO to improve search visibility and organic discovery.
- Drove about 30x the usual daily traffic on launch day (a month's worth in one day) through the redesign and SEO work; used analytics to measure the impact and guide next steps.

PROJECTS

AyurIP-Sahayak | Multilingual RAG Assistant ayur-ip-sahayak.vercel.app
Python, FastAPI, React, Google Gemini API, NumPy, Docker, GitHub Actions | github.com/rudraksh01-tech/AyurIP-Sahayak

- Built a RAG assistant that answers questions on Ayurveda, traditional knowledge and Indian IP law in English, Hindi, Kannada and Hinglish, citing the exact source page for every claim and streaming the first words in under 2 seconds.
- Designed the retrieval pipeline: section-aware PDF chunking, Gemini embeddings (768 dimensions), NumPy cosine-similarity vector search, and BM25 with Reciprocal Rank Fusion hybrid search written from scratch.
- Benchmarked retrieval on 38 labelled questions: Hit@1 0.95 and MRR 0.97, against 0.47 and 0.60 for a BM25 keyword baseline; kept dense search as the default after measuring that hybrid search lowered MRR to 0.93.
- Added an LLM-as-a-judge answer evaluation with a model from a different family: 0.99 faithfulness (166 of 167 claims supported), 0.98 citation precision, 5 of 5 off-topic questions refused, and 15 of 15 planted errors caught.
- Fixed an out-of-memory deployment by swapping a local PyTorch model for an embeddings API; added Docker, 47 tests and CI.

NestFinder | Agentic AI Assistant for Flat Search nestfinder-woad.vercel.app
Node.js, Express.js, React, MongoDB, Groq API, Langfuse, OpenTelemetry | github.com/rudraksh01-tech/nestfinder

- Developed an agentic AI assistant with tool calling from scratch (no framework): 5 tools, streamed responses, a repair loop for rejected tool calls and automatic fallback to a second model on rate limits.
- Kept decisions deterministic: a unit-tested rules engine computes match scores, fair-rent medians and distances, while the LLM handles language and tool choice, so it cannot invent numbers.
- Wrote a 20-case agent evaluation covering Hindi, Hinglish and prompt-injection inputs; fixed a schema bug it exposed, raising tool-choice accuracy from 18/20 to 20/20.
- Traced every LLM call with OpenTelemetry and Langfuse, masking phone numbers before export; added JWT auth in httpOnly cookies, bcrypt, rate limiting and MongoDB geospatial queries; deployed on Vercel and Render with CI.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

AI / LLM: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Tool Calling, Embeddings, Semantic Search, Vector Search, Hybrid Search (BM25, RRF), LLM Evaluation (LLM-as-a-Judge), Structured Outputs (JSON Schema), Prompt Engineering, Google Gemini API, Groq API

Backend: FastAPI, Node.js, Express.js, REST APIs, MongoDB, JWT Authentication

Frontend: React, Vite, Tailwind CSS, HTML, CSS

DevOps & Tools: Docker, GitHub Actions (CI/CD), Git, GitHub, Linux, pytest, Vercel, Render, Langfuse, OpenTelemetry

Core CS: Object-Oriented Programming (OOP), Operating Systems, DBMS, Computer Networks